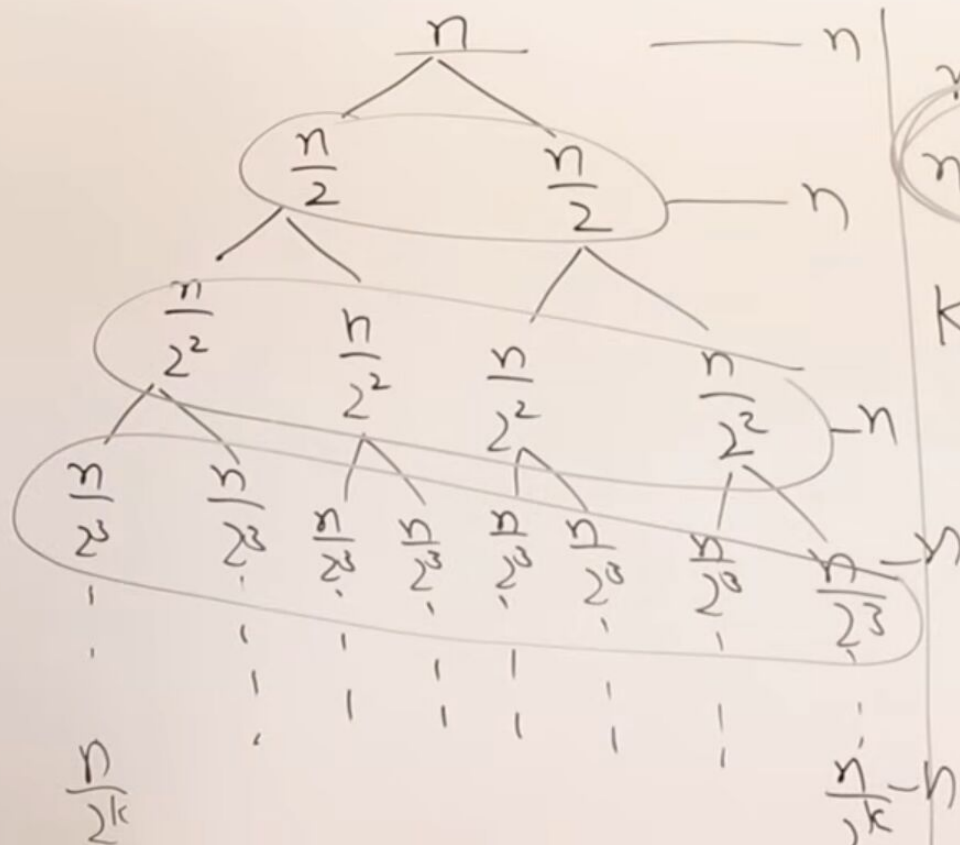


Recurrence Relation

$$T(n) = \begin{cases} 1 & n=1 \\ 2T(n/2) + n & n>1 \end{cases}$$

Assume $\frac{n}{2^k} = 1$
 $n = 2^k \quad k = \log n$



$n \log n$

k

$T(n)$ — void Test(int n)
 {

if (n > 1)
 {

for (i = 0; i < n; i++)
 {
 stmt;
 }

$T(n/2)$ — Test(n/2);
 $T(n/2)$ — Test(n/2);
 }

$T(n) = 2T(n/2) + n \quad O(n \log n)$